

JNEOPALLIUM · CYBERSECURITY

Your Network Deserves an Immune System

Temporal threat correlation that catches the attack, not just the alert

Demo 06 · cybersecurity-temporal-threat-correlator · ADVISORY-safe · BSD-3

01 The attacker is already inside — for weeks

Detection is a race against dwell time, and most teams are losing it.

- ▶ The average intruder dwells for **weeks** before anyone notices — long enough to steal credentials, spread, and exfiltrate.
- ▶ Every wasted hour widens the blast radius: more hosts, more data, more cost.
- ▶ By the time a single alert is convincing on its own, **the damage is already done**.
- ▶ The question that matters is not "is this event bad?" — it is "is an attack unfolding **right now**?"

02 Why today's tools keep missing it

One event at a time

- Scores each login, file, or packet ****in isolation****
- ****Low-and-slow**** theft never trips a threshold
- Drowns analysts in ****thousands of daily alerts****
- Can't tell a 3 a.m. finance-server script from routine maintenance

What's actually needed

- Connect events ****across time and sources****
- Recognise the ****shape of an attack chain****
- One evidenced advisory, ****not a thousand alarms****
- Apply context ****without erasing the evidence****

03 The big idea: a digital immune system

We borrowed the best intrusion-detection system on Earth — biology's.

- ▶ **Fast first-responders** match known threats instantly (like macrophages).
- ▶ **Adaptive learners** model what's normal for each user and host (like T-cells).
- ▶ **Immune memory** recalls past campaigns so new ones are caught faster (like memory B-cells).
- ▶ **Self-tolerance** is a hard, un-overridable rule never to harm your own critical systems.
- ▶ **Inflammation always resolves** — every containment expires automatically. Nothing is permanent.

04 How it works — in three moves

- ▶ **1 • Translate.** Every login, process, DNS lookup, network flow, and threat-intel tip becomes a typed signal in one common language.
- ▶ **2 • Correlate.** The engine connects those signals across overlapping time windows — seconds, minutes, and the full 30–120 minute incident window — using **event time**, even when data arrives late or out of order.
- ▶ **3 • Advise.** It outputs a single recommendation with a probability, a response band, and a full evidence trail you can explain to any auditor.

05 It sees the chain, not the link

A real intrusion is an ordered story. We read the whole story.

- ▶ unusual login → encoded PowerShell → lateral movement → DNS → C2 beaconing → data exfiltration
- ▶ Each step alone looks almost normal. **In sequence, within the window, it's an intrusion.**
- ▶ Ordered transitions score far higher than the same events scattered at random.
- ▶ And the **low-and-slow leak** — a few megabytes an hour — finally becomes visible when correlated over the incident window.

06 Three streams. Three correct calls. Zero blocks.

The exact scenario one-event tools get wrong — handled right.

- ▶ **The real attack** is raised as a temporal threat advisory, with learning frozen so it can't poison "normal."
- ▶ **Planned maintenance** is calmed by context — but its evidence is still recorded for audit.
- ▶ **The quiet data leak** is surfaced by correlation, not by a single threshold.
- ▶ In every case the output is an **advisory with evidence** — never a silent, business-breaking block.

07 Why we win

What competitors structurally cannot copy overnight.

- ▶ **Temporal correlation by design** — built around the attack chain, not a single-row classifier.
- ▶ **Safe by construction** — ADVISORY by default; hard safety gates are fixed, never learned.
- ▶ **Glass-box, not black-box** — training emits the actual deployable network; every weight, gate, and threshold is inspectable, and every advisory carries its evidence lineage.
- ▶ **Train-to-deploy in one step** — what you train is the exact network you run; no model-translation gap.
- ▶ **Multi-source from day one** — authentication, process, DNS, flow, threat-intel, asset, and maintenance context, fused.
- ▶ **Built on published science** — the Jneopallium framework, peer-reviewed in IJSR (2024).

08 It cannot take your business offline

The number-one objection to autonomous security — answered structurally.

- ▶ **Advisory by default.** It recommends; humans decide. Enforcement needs a separate, approved safety case.
- ▶ **Critical assets are protected by a fixed gate** that no runtime signal — and no attacker in the data path — can override.
- ▶ **Baseline freezes during attacks**, defeating slow "baseline-poisoning" tricks.
- ▶ **Every quarantine expires automatically.** Being wrong is, by design, safe.

09 Proof: the pipeline is real

Perfect separation on the reference corpus — machinery proven, ready for your data.

1.0

Precision & recall
(reference corpus)

0.0

False-positive rate

100 GiB

Logical-corpus scale
proven on a workstation

ADVISORY

Safety ceiling,
never exceeded

10 Honest about the evidence

Because a vendor who hides this can't be trusted with your network.

- ▶ Those perfect scores are **pipeline evidence** on a deterministic reference corpus — they prove the system is wired correctly end to end.
- ▶ Real-world accuracy is **earned on your data**: LANL, ToN_IoT, OpTC, CIC/CSE-CIC, UNSW-NB15, and controlled CALDERA campaigns plug straight in.
- ▶ The training pipeline already enforces leakage-safe splits, calibration, and an F1 quality gate.
- ▶ **We lead with this caveat on purpose.** The honesty is part of the product.

11 Fits where you already are

Built to slot into the enterprise, not replace it.

- ▶ **Kafka-native** event stream design — point it at your existing telemetry bus.
- ▶ Deploys **local, HTTP cluster, or gRPC** (FPGA-capable) — laptop pilot to enterprise scale.
- ▶ **Shadow-mode rollout:** run silently, compare against your incident tickets, then promote to operator-visible advisory.
- ▶ Detectors are **swappable behind interfaces** — bring your own signature engine or anomaly model.
- ▶ **Open source, BSD-3-Clause** — no lock-in, full auditability.

12 Where it pays off

Best fit

- SOCs drowning in alert fatigue
- Enterprises chasing **dwelt-time** reduction
- Regulated sectors needing **explainable** verdicts
- MSSPs needing multi-tenant, auditable triage

What you get

- Fewer, **higher-quality** advisories
- Earlier detection of **multi-step** attacks
- A full **evidence trail** for every call
- A safe path from **shadow** → advisory

13 Where it's going

A clear roadmap on a proven foundation.

- ▶ Validation on external multi-source datasets to convert pipeline evidence into **field accuracy**.
- ▶ Larger sequence models (**GRU / TCN / transformer**) behind the same transparent interface.
- ▶ Production Kafka bridge, containerised cluster deployment, and hosted model packaging.
- ▶ Optional, non-blocking **LLM knowledge-base** advisory for analyst context — never in the critical path.

Let's catch the attack — not just the alert

A digital immune system for your network: temporal, transparent, and safe by construction.

Pilot it in shadow mode on your own telemetry in weeks, not quarters.

Open source today. Production-ready advisory on your roadmap.

Let's run a pilot on your data.